

Java Regex

Pattern Matching & Text Validation

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What is Regex?

A **Regular Expression** (Regex) is a sequence of characters that forms a search pattern. When you search for data in a text, you can use this search pattern to describe what you are looking for.

In Java, the **java.util.regex** package is used for pattern matching. It includes three main classes:

- **Pattern Class:** Defines a pattern (to be used in a search).
- **Matcher Class:** Used to search for the pattern.
- **PatternSyntaxException:** Indicates a syntax error in a regex pattern.

1. Basic Regex Example

Here is how to check if a specific word exists in a string regardless of case sensitivity:

```
import java.util.regex.Matcher;
import java.util.regex.Pattern;

public class Main {
    public static void main(String[] args) {
        Pattern pattern = Pattern.compile("codehive",
Pattern.CASE_INSENSITIVE);
        Matcher matcher = pattern.matcher("Visit CodeHive today!");
        boolean matchFound = matcher.find();

        if(matchFound) {
            System.out.println("Match found");
        } else {
            System.out.println("Match not found");
        }
    }
}
```

2. Regex Pattern Reference

Patterns define the "rules" of your search. Here are the most common metacharacters utilized in Java:

Pattern	Description
[abc]	Find one character from the options inside the brackets.
[^abc]	Find one character NOT inside the brackets.
[0-9]	Find one character from the range 0 to 9.
\d	Find a digit (short for [0-9]).
\s	Find a whitespace character.
\b	Find a match at the beginning/end of a word.
.	Find any single character.

Note: In Java, backslashes must be escaped. So, instead of \d, you write \\d.

3. Quantifiers

Quantifiers define how many times a character (or group) must appear for a match to occur.

Quantifier	Description
<code>n+</code>	Matches any string that contains at least one <i>n</i> .
<code>n*</code>	Matches any string that contains zero or more occurrences of <i>n</i> .
<code>n?</code>	Matches any string that contains zero or one occurrence of <i>n</i> .
<code>n{x}</code>	Matches any string that contains a sequence of exactly <i>x</i> <i>n</i> 's.

```
// Example: Validate a 3-digit number
Pattern p = Pattern.compile("^\\d{3}$");
Matcher m = p.matcher("123");
System.out.println(m.matches()); // true
```

Summary Checklist

- ✓ Regex is used for searching, editing, or manipulating text.
- ✓ Use `Pattern.compile()` to define your search rules.
- ✓ Use `Matcher` to perform the actual search.
- ✓ Remember to escape backslashes (`\\`) in Java strings.

Mastering Patterns with

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Precision in Every String.